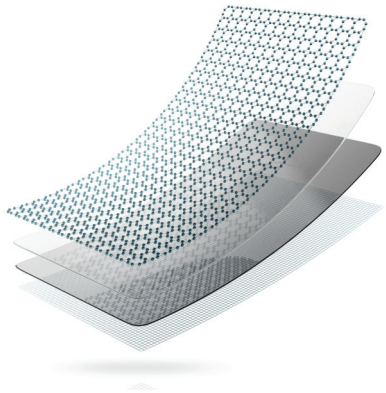


Surface of Silicon

DO YOU KNOW?

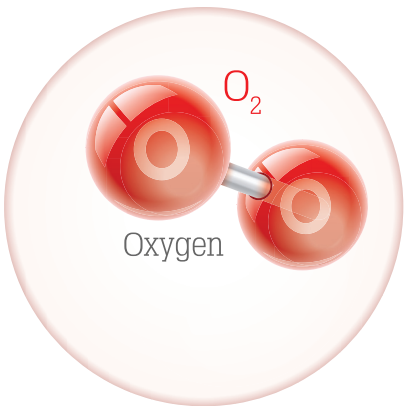
The most abundant atom in the universe is the hydrogen atom. Nearly 74% of the atoms in the universe are hydrogen atoms. However, three most abundant atoms on the Earth are iron, oxygen and silicon.

## 3.2 Molecules

When an atom combines with another atom (or atoms) and forms a compound, it is called as molecule. *A molecule is made up of two or more atoms chemically combined.*

OX397

Oxygen gas in the air that we breathe is made up of two oxygen atoms chemically combined.

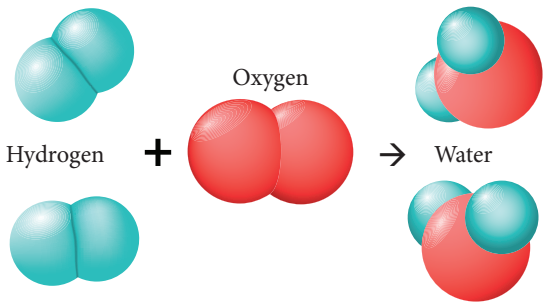



Ozone is a substance that is made up of three oxygen atoms chemically combined.




An atom of oxygen (O) and two atoms of hydrogen (H) combine to form a molecule of water ($H_2O$).

$$2H_2 + O_2 \rightarrow 2H_2O$$




Formation of water molecule

Molecules exhibit the properties of matter and also have individual existence. A molecule can be formed by the same or different kinds of atoms.

Molecules can be classified as below.

- A molecule which contains only one atom is called monatomic molecule (Inert gases).
- A molecule which contains two atoms is called diatomic molecule (Oxygen, Nitric oxide, Hydrogen, etc.).
- A molecule containing three atoms is called a triatomic molecule (Ozone, Sulphur dioxide, Carbon dioxide, etc.).
- A molecule containing more than three atoms is known as polyatomic molecule (Phosphate, Sulphur, etc.).